

Scope

FROM SEPA-9Ed:

Software scope describes the functions and features that are to be delivered to end users; the data that are input and output; the "content" that is presented to users as a consequence of using the software; and the performance, constraints, interfaces, and reliability that bound the system. Scope can be defined by developing a set of use cases that is developed with the end users.

Functions described in the use cases are evaluated and in some cases refined to provide more detail prior to the beginning of estimation. Because both cost and schedule estimates are functionally oriented, some degree of decomposition is often useful. Performance considerations often constrain processing and response-time requirements. Once scope has been identified (with the concurrence of the customer), it is reasonable to ask: "Can we build software to meet this scope? Is the project feasible?" All too often, software engineers rush past these questions (or are pushed past them by impatient managers or other stakeholders), only to become mired in a project that is doomed from the onset. You must try to determine if the system can be created using available technology, dollars, time, and other resources. Project feasibility is important, but a consideration of business need is even more important. It does no good to build a high-tech system or product that no one wants.

The first software project management activity is the determination of software scope. Scope is defined by answering the following questions:

- Context. How does the software to be built fit into a larger system, product, or business context, and what constraints are imposed as a result of the context?
- Information objectives. What customer-visible data objects are produced as output from the software? What data objects are required for input?+
- Function and performance. What function does the software perform to transform input data into output? Are any special performance characteristics to be addressed?

Software project scope must be unambiguous and understandable at the management and technical levels. A statement of software scope must be bounded. That is, quantitative data (e.g., number of simultaneous users, size of mailing list, maximum allowable response time) are stated explicitly, constraints and/or limitations (e.g., product cost restricts memory size) are noted, and mitigating factors (e.g., desired algorithms are well understood and available in Java) are described. Even in the

most fluid situations, the number of prototypes needs to be considered and the scope of the first prototype needs to be set.

Bound the Scope

Bounding the scope involves setting clear, measurable limits on the project.

- **Define the project's scopes and constraints.**
 - **Documentation Requirement:** A section titled "Project Scopes and Constraints" is a mandatory component of both the Senior Project Proposal Draft and the IT483 Report.
 - **Core Definition:** The act of binding the scope involves describing:
 - The functions and features to be delivered.
 - The data that are input and output.
 - The content presented to users.
 - The performance, constraints, interfaces, and reliability that bind the system.
 - **Essential Characteristics:** A well-defined statement of software scope must possess the following qualities:
 - It must be **bound**, meaning it has clear and explicit limits.
 - It must be **unambiguous and understandable** at both the management and technical levels.
 - **Specific Details to Include:** To properly bound the scope and define constraints, the documentation should:
 - State quantitative data explicitly for special performance characteristics to be addressed, such as time, speed, volume (the number of simultaneous users), schedule constraints, or maximum allowable response time.
 - Note: Performance considerations often place constraints on processing and response-time requirements. The scope must explicitly state quantitative data. The "User Requirements" section of the report must specify these performance requirements, including any time, speed, volume, or schedule constraints the system must fulfill.
 - Note specific constraints and/or limitations, such as how product cost restricts memory size.
 - Describe any mitigating factors, for example, noting that desired algorithms are well understood and available.
 - Include a description of constraints related to time, funds, skills, technology, and external factors.

- Specify control requirements, including access, user roles, or audit needs.
 - This is a specific type of user requirement that must be documented in the report. The control requirements section specifies needs related to access, users, roles, or auditing. For IT Senior Projects that are Management Information Systems, a minimum security requirement is the inclusion of user accounts and roles.

Example

Project Title

Web-Based Equipment Lending System for IT Lab

Project Scopes and Constraints



Note

A statement of software scope must be bound. This section explicitly defines the boundaries, performance, and limitations of the Equipment Lending System.

In-Scope:

- **Functions and Features:** The system will provide functions for managing equipment inventory, user accounts, and lending requests. It will include features for adding, updating, and deleting equipment records, as well as tracking the status of each item (e.g., available, on loan, under maintenance).
- **Data and Complexity:** The system will manage a minimum of 7 data-objects (e.g., Equipment, User, Role, Loan Request, Category, Manufacturer, Location). The user interface will consist of a minimum of 30 use cases to handle all management functions.
- **Users:** The system will support two user roles: "Student" and "Lab Administrator".

Out-of-Scope:

- The system will **not** process fines or payments for late or damaged equipment.
- The system will **not** integrate with the university's main student registration database.
- The system will **not** include a public-facing page for browsing equipment; access is restricted to authenticated users.

Quantitative Performance Requirements:

- The system must support up to 20 simultaneous users, based on the potential maximum number of staff in IT Services Department.
- The maximum allowable response time for searching the equipment inventory must be under 2 seconds, based on Shneiderman's (1984) and Miller's (1968) theory about human short-term memory and information processing.
- The system must be capable of managing a list of up to 1,000 unique equipment items.

Constraints and Limitations:

- **Technical:** The project is constrained to using the university-provided remote database server and institutional GitHub repository.
- **Financial:** The project has a zero-dollar budget, which restricts the use of paid third-party APIs or software components.
- **Schedule:** The project is strictly constrained by the official Senior Project Schedule, with deliverables due in week 15 and week 33.